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PFAS and heavy metals mixture *in vitro*: Insights into synergistic toxicity across multiple cell lines

Mia Sands [a,b], Arshveer Sachdeva [a], Laura Bukavina [f], Joseph Irudayaraj [a,b,c,d,e,*]

[a] Department of Bioengineering, University of Illinois, Urbana-Champaign, Urbana, IL, 61801, USA
[b] Biomedical Research Center, Mills Breast Cancer Institute, Carle Foundation Hospital, Urbana, IL, 61801, USA
[c] Department of Comparative Biosciences, College of Veterinary Medicine, University of Illinois Urbana-Champaign, Urbana, IL, 61801, USA
[d] Carl Woese Institute for Genomic Biology, University of Illinois, Urbana-Champaign, Urbana, IL, 61801, USA
[e] Cancer Center at Illinois, Beckman Institute, Urbana, IL, 61801, USA
[f] Cleveland Clinic Glickman Urological Institute, Cleveland, OH, 44195, USA

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ABSTRACT

Poly- and perfluoroalkyl substances (PFAS) are ubiquitous environmental contaminants originating from industrial processes. While the individual toxicities of PFAS and heavy metals have been extensively studied, their co-occurrence in the environment raises concerns on potential interactions that may amplify their effects on human health. Specifically, the manner in which heavy metals influence PFAS absorption, bioavailability, and combined toxicity remain poorly understood. This knowledge gap limits our ability to assess the true health risks associated with real-world co-exposures to these contaminants. To address this gap, our study investigates the mixture effects of PFAS and heavy metals on human cell viability, proliferation, migration, and DNA damage across five human cell lines (HepG2, A498, A549, A431, and PC3) at human-relevant exposure levels. In our study, 1 μM PFOA and GenX, along with 0.01 μM lead and arsenic, were chosen to represent human exposure levels. Our results revealed cell-type-specific responses, with mostly synergistic interactions observed across different cell lines. HepG2 and A549 cells exhibited impaired cell proliferation, increased DNA damage, and reduced cell migration upon mixture exposure, while A498 and PC3 cells showed increased cell proliferation, with less pronounced effects on DNA damage and migration. These findings suggest that PFAS and heavy metal mixtures can impact cellular functions through cell-type and combination-specific toxicological pathways. This study provides new insights into the complex interactions between PFAS and heavy metals, highlighting the need for further research to better understand the underlying mechanisms and its impact on health risks associated with co-exposure to these environmental contaminants.

1. Introduction

Per and poly-fluoroalkyl substances (PFAS) and heavy metals are both ubiquitous environmental contaminants that can enter the human body through inhalation or ingestion and accumulate in various tissues in the body (Evich et al., 2022; Fulke et al., 2024; Glüge et al., 2020; Witkowska et al., 2021). PFAS are fluorinated carbon chains, identified by the Toxic Substances Control Act (TSCA) and have been widely used in industrial and consumer products due to their effectiveness as surfactants or surface protectors (Glüge et al., 2020). Perfluorooctanoic acid (PFOA) is one of the most widely used PFAS chemicals and has been detected in human serum in various locations around the world (Bartell and Vieira, 2021; Cheng et al., 2022). Similarly, heavy metals can be found in water both naturally and as a result of industrial activities such as mining, burning of fossil fuels, and the use of fertilizers and pesticides (Giroux et al., 2024; Parida and Patel, 2023). Lead and arsenic are two of the high-priority heavy metal contaminants due to their wide presence and high toxicity (Ali et al., 2019).

PFOA has been found to accumulate in the organs involved in xenobiotic metabolism such as liver and kidneys and reproductive systems and has been associated with nephrotoxicity, neurotoxicity, genotoxicity, immunotoxicity, and hepatotoxicity (Li et al., 2017; Liu et al., 2023; Rashid et al., 2020; Tarapore and Ouyang, 2021; Wen et al., 2020, 2022). Due to its harmful environmental impacts, in recent years, in response, manufacturers have shifted towards the adoption of short-chain and ultra-short-chain PFAS, along with other fluorinated

* Corresponding author at: Biomedical Research Center, 3rd Floor Mills Breast Cancer Institute, 502 N. Busey Ave, Urbana, IL, 61801, USA.
E-mail address: jirudaya@illinois.edu (J. Irudayaraj).

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alternatives such as perfluoropolyethers, in commercial production (Brunn et al., 2023; Hamid et al., 2023; Li et al., 2020b; Smaili and Ng, 2023). For example, Hexafluoropropylene oxide dimer acid (GenX) has been identified as a substitute for perfluorooctanoic acid (PFOA) to mitigate environmental and health risks (Heidari et al., 2021; Hopkins et al., 2018). Despite being introduced as a safer alternative to PFOA, GenX has also been linked to various toxicity (Gebbink and van Leeuwen, 2020; Heidari et al., 2021; Rashid et al., 2023; Sheng et al., 2018; Wen et al., 2020).

Lead primarily enters the body through inhalation or ingestion, and circulates in the form of soluble salts or protein complexes (Hanfi et al., 2019). Once in the body, lead is stored in the bones and teeth but can also damage the liver, the kidneys, reproductive organs, nerve systems and brain (Boskabady et al., 2018; Collin et al., 2022; Hanfi et al., 2019). Lead poisoning and exposure can result in proximal tubular dysfunction, reduced glomerular filtration, liver disease, neurological disease, and heart disease (Collin et al., 2022; Lee et al., 2019; Nakhaee et al., 2019). Additionally, it can cause lung damage such as dyspnea and pleural effusion, as well as damage to the skin (Samarghandian et al., 2021). In vitro studies have found that lead exposure can result in oxidative stress, DNA methylation, endocrine disruption, and cell membrane destruction (Wani et al., 2016). The permissible blood concentration for lead is 0.01 mg/L (Ali et al., 2024), however, occupational exposure can reach up to a mean of 0.11 mg/L (Mohammadyan et al., 2019). In addition, an NRDC report from 2021 found that 186 million people in the United States drank water with lead contamination exceeding the limit of 1 parts-per-billion recommended by the American Academy of Pediatrics, and 61 million people consumed water that exceeded 5 ppb (parts-per-billion) (Fedinick, 2021). Similarly, arsenic can enter the body through contaminated food and water, and in 2003, a WHO report estimated 140 million people were exposed to high levels of inorganic arsenic (Cordier et al., 2021; Folesani et al., 2023). The effects of arsenic ingestion are similar to that of lead and can result in liver cancer, cardiac disorders, kidney damage such as proximal tube degeneration, and neurological damage (Cordier et al., 2021). Inhalation of arsenic can induce lung cancer, and dermal exposure can cause skin cancer (Fatoki and Badmus, 2022; Rajiv et al., 2023). It has been reported that the toxicity of arsenic is primarily induced due to oxidative stress (Fatoki and Badmus, 2022).

Recent studies have increasingly evaluated the effect of PFAS and heavy metal mixtures on the environment. It has been found that sorption of PFAS by soil is increased in the presence of heavy metal ions, which can further expose populations to these contaminants by increasing mobility of PFAS throughout the environment (Cai et al., 2023, 2022; Campos-Pereira et al., 2023; Kookana et al., 2023). Clinically, the recent rise in chronic diseases such as cardiovascular disorders and cancers has been attributed to increased exposure to environmental contaminants like heavy metals and PFAS, with mixtures which include lead showing the greatest associated risk for cardiovascular disease (Boafo et al., 2023; Furman et al., 2019; Odediran and Obeng-Gyasi, 2024). Interestingly, in the context of depression, PFAS and heavy metals appear to exhibit antagonistic interactions, as PFAS exposure has been reported to counteract heavy-metal-induced depressive effects (Tang et al., 2024). Other clinical studies have shown that co-exposure to PFAS and heavy metals is associated with hepatic fibrosis, osteoporosis, impaired kidney health, and pregnancy-loss (Fang et al., 2024; Haruna and Obeng-Gyasi, 2024; Liang et al., 2023; Wang et al., 2025; Y. Wei et al., 2024,Z. 2024). Despite this emerging evidence, the combined cellular health impacts and underlying mechanisms of PFAS–heavy metal co-exposures remain largely unexplored, representing a critical research gap. Prior exposures to persistent chemicals may lower human tolerance to additional environmental stressors, underscoring the need to move beyond single-chemical toxicology.

To address this gap, the present study systematically evaluates the interactive effects of co-exposure to PFAS and heavy metals compared to their individual toxicities in vitro. Specifically, we aim to characterize whether these mixtures result in additive effects, synergistic effects (toxicity greater than expected from individual exposures), or antagonistic effects (toxicity less than expected). We assess not only general cytotoxicity but also key functional outcomes, including cell proliferation, migration, and DNA damage. To model organ-specific responses, we used cell lines derived from the kidney, liver, lung, prostate, and epidermis—tissues commonly impacted by PFAS and heavy metal exposure. Although previous studies have established that PFAS and heavy metals independently induce oxidative stress, genotoxicity, and other forms of cellular damage (Lopes et al., 2016; Sanchez et al., 2017; Wen et al., 2020; Wielsøe et al., 2015), their mixture effects remain poorly characterized. We hypothesize that co-exposure to PFAS and heavy metals leads to synergistic interactions, resulting in altered cellular function and increased damage across multiple biological endpoints.

2. Materials and methods

2.1. *Chemicals and test reagents*

The ammonium salt of Perfluorooctanoic acid (PFOA) (CAS No. 335-67-1; ≥95.0 % purity) was purchased from Sigma-Aldrich (St. Louis, MO, USA). The Undecafluoro-2-methyl-3-oxahexanoic acid form of GenX (CAS No. 13,252-13-6) of 97 % purity was purchased from Synquest labs (Alachua, Florida, USA). Lead nitrate (CAS No. 228,621; ≥99.0 % purity) and Sodium (meta)arsenite (CAS S7400; ≥90.0 % purity) were also purchased from Sigma-Aldrich. For all cell exposure experiments, PFOA and GenX stock solutions were prepared by dissolving them in dimethyl sulfoxide (DMSO; CAS No. D8418) from Sigma-Aldrich to obtain a stock solution of 0.02 M. Arsenic and Lead stock solutions were prepared by dissolving them in sterile double-distilled water to obtain a stock solution of 0.02 M. The chemical structure of PFOA and GenX are shown in **Supplementary Fig. 1**.

2.2. *Cell culture*

The HepG2 (ATCC HB-8065), A498 (ATCC HTB-44), A549 (ATCC CCL-185), A431 (ATCC CCL-185), and PC3 (ATCC CRL-1435) cell lines were obtained from the Cancer Center at Illinois, University of Illinois Urbana-Champaign. HepG2, A549, and A431 cells were cultured in Dulbecco's Modified Eagle's Medium (ATCC 30–2002; Manassas, VA, USA) supplemented with 10 % fetal bovine serum (FBS, (Gibco™ 10,082,147; Thermo Fisher Scientific; Waltham, MA, USA) and 1 % Penicillin Streptomycin Solution (REF 30–002-Cl; Corning, NY, USA). A498 was cultured in Eagle's Minimum Essential Medium (ATCC 30–2003; Manassas, VA, USA) supplemented with 10 % FBS and 1 % Penicillin Streptomycin Solution. PC3 and LNCAP were cultured in RPMI (Corning 10-040-CV; NY, USA) supplemented with 10 % FBS and 1 % Penicillin Streptomycin Solution. Culture conditions for all cells were at 37 °C in a humidified 5 % CO_2 atmosphere. For single chemical and mixture treatments, cells were seeded into six-well plates and cultured until they reached 30–40 % confluence (density around 5 × 10^5). Afterward, the cells were treated with the corresponding chemicals for 48 h (A549 were treated for 24 h due to fast proliferation) and subsequently harvested for downstream experiments. All experiments were performed independently at least in triplicate.

2.3. *3-[4,5-dimethylthiazol-2-yl]-2,5 diphenyl tetrazolium bromide (MTT) assay*

Cell growth and proliferation assay were measured with the MTT assay kit (Invitrogen, Cat. No. M6494). Cells were plated into a 96-well microplate (Corning Incorporated, New York, USA) and incubated at 37°C in 5 % CO2. Each data point represents measurement from three replicate wells. The cells were cultured to 30–40 % confluence (density around 2 × 10^5) before corresponding treatments and cultured for another 24 h or 48 h to reach a final confluence of 60–80 % (density around 3.5×10^5). For HepG2, A498, A431, and PC3, we performed 48 h treatments. For A549 cell line, only 24 h treatments were considered due to the rapid cell growth. After exposure to chemicals, 10 µl of MTT reagent was added to each well with 90 µl of serum-free medium and incubated for another 4 h at 37 °C. DMSO was then added to solubilize the formazan crystals, and absorbance at 570 nm was measured after a 20-minute using a microplate reader (Synergy HT, BioTek; Winooski, VT, USA). Cell viability was determined as a percentage of absorbance values at different doses in relation to their vehicle control and assessed from: $[OD_{treatment} - OD_{blank}]/[OD_{vehicle\ control} - OD_{blank}] \times 100\ \%$.

2.4. Single chemical effects on cell viability

HepG2, A498, A549, A431, and PC3 cell lines were exposed to ten concentrations of each PFAS compound (1, 5, 10, 20, 40, 80, 120, 160, 200, and 240 µM), along with solvent-free media as a negative control and a 0.1 % DMSO solvent control. Similarly, these cell lines were exposed to ten concentrations of each heavy metal compound (0.01, 0.025, 0.05, 0.1, 1, 2.5, 5, 10, 15, and 20 µM), with a negative control in solvent-free media. Each experiment included at least four biological replicates. The EC_{10} values for each compound in each cell line were calculated using linear models (Escher et al., 2018; Ríos-Bonilla et al., 2024) in R (version 4.4.2), focusing on effects below 30 %. To ensure realistic exposure assessments, concentration-response curves (CRCs) were fitted using linear models for low-effect levels (<30 % effect). At these levels, most monotonic CRCs from biochemical assays approximate a linear relationship with an intercept at zero (Escher et al., 2018). The EC_{10} and their standard errors were calculated using Eqs. (1) and (2).

$$EC_{10} = \frac{10\%}{slope} \tag{1}$$

$$SE(EC_{10}) = \frac{10\%}{slope^2} \times SE(slope) \tag{2}$$

2.5. Mixture effects on cell viability

For each cell line, four binary mixture experiments were conducted to assess cytotoxicity: PFOA-Pb, PFOA-As, GenX-Pb, and GenX-As. Mixture testing was based on the toxic unit (TU) concept, where a TU is defined as the concentration of a chemical (c) divided by a toxicity threshold (in this case, the EC_{10} value), as shown in Eq. (1):

$$TU = \frac{c}{EC_{10}} \tag{3}$$

Exposure scenarios were designed under the assumption of additivity, following a fixed factorial experimental design. Mixtures were tested at three TU dose ratios (3:1, 1:1, and 1:3) and five total TU levels ($\sum TU$=0.01, 0.1, 0.5, 1, 2).

For low-effect levels (< 30 % cytotoxicity inhibition), the EC_{10} of a concentration-additive mixture (EC_{10} (CA) was predicted by Eq. (2) assuming n components i are present in fractions p_i such as $\sum p_i = 1$ (Ríos-Bonilla et al., 2024; Tanaka and Tada, 2017) **(Supplementary Information – Mixture Calculations)**.

$$EC_{10}(CA) = \frac{1}{\sum_{i=1}^{n} \frac{p_i \times slope_i}{10\%}} = \frac{10\%}{\sum_{i=1}^{n} p_i \times slope_i} \tag{4}$$

$$SE(slope_{CA}) = \sqrt{\sum_{i=1}^{n} {p_i}^2 \times SE(slope_i)^2} \tag{5}$$

To determine the interaction between PFAS and heavy metals, the observed effects of co-exposure were compared to the expected additive effects. Interactions were classified as synergistic if the observed combined effect significantly exceeded the predicted additive effect, or antagonistic if it was significantly lower. Effects not significantly different from the predicted value were considered additive. This classification approach is consistent with established methods in mixture toxicology (Berenbaum, 1977; Olszowy-Tomczyk, 2020; Warne and Hawker, 1995).

2.6. Colony formation

Untreated cells, as well as those treated with single chemicals or chemical mixtures for 48 h, were seeded into 24-well plates at a density of 200 cells per well and cultured for one week. Following incubation, the cells were fixed with 4 % paraformaldehyde for 15 min and stained with 0.5 % crystal violet solution for approximately 5 min. Colonies containing 50 or more cells were counted as individual colonies. Data was collected and quantified using ImageJ software.

2.7. Scratch assay

Untreated, as well as those treated with single chemicals or chemical mixtures for 48 hours, were pre-seeded into a 24-well plate. Once the cells reached 80 %–90 % confluency, controlled scratch wounds were created using a sterile 200 µL pipette tip. The culture medium was then replaced with medium containing 1 % FBS. Wound healing was monitored after 24 h under a microscope and quantified using ImageJ software.

2.8. DNA damage assessment

DNA damage was evaluated using the OxiSelect Comet Assay Kit (STA-850) with modifications. Untreated and treated cells were harvested and resuspended in 1 mL of phosphate-buffered saline (PBS). Glass slides were prepared with a two-layer agarose gel and subjected to electrophoresis, followed by staining with 1 × Vista Green DNA dye (provided with the kit). Fluorescent images were captured immediately using the fluorescence light microscope (Zeiss, Germany). The extent of DNA damage was quantified using CASP software, with the olive tail moment serving as the metric for analysis (Hellman et al., 1995). This parameter was chosen because it combines both the distance of DNA migration (tail length) and the fraction of DNA present in the tail, offering a more integrated and sensitive measure of DNA strand breaks. The Olive tail moment is calculated as: [Olive moment= (tail mean-head mean) ×% of DNA in the tail], providing a more comprehensive measure of DNA strand breaks (Kumaravel et al., 2009; Mozaffarieh et al., 2008; Olive et al., 1990). Positive controls were included by treating cells with 50 µM H_2O_2 on ice for 30 min. For each sample, 100–200 random comets were scored using the same software.

2.9. Statistical analysis

Statistical analysis of data was performed with GraphPad Prism 9.5 Software. A two-tailed unpaired Student's test was selected to assess difference between means of different dosage treatment groups. Analysis of variance (ANOVA) followed by Tukey's Honestly Significant Difference (HSD) test was employed to evaluate potential statistically significant differences among the means of multiple treatment groups. A significance level of $P < 0.05$ was considered as indicating statistical significance.

3. Results

3.1. Single chemical exposures across human cell lines

The effects of PFAS and heavy metals on cell viability were evaluated in five human cell lines representing different tissues: liver (HepG2), kidney (A498), lung (A549), epidermis (A431), and prostate (PC3).

PFAS exposures (0–240 μM) included long-chain PFOA and the short-chain replacement GenX, while heavy metal exposures (0.01–20 μM) included arsenic and lead. Cells were treated for 48 h, except for A549 cells, which were treated for 24 h due to their rapid proliferation. No significant difference was observed between the untreated control and the vehicle control containing 0.1 % DMSO **(Supplementary Fig. 2)**. Effective concentrations of EC_{10} values were determined from a linear evaluation of concentration-response curves for each chemical in each cell line (Table 1). Doses causing effects greater than 30 % were excluded from the analysis. In PC3 cells, single heavy metal treatments resulted in increased cell viability, producing a negative slope in the dose–response curve (Table 1). To account for this and reflect the antagonistic effect, the absolute value of the slope was used in subsequent calculations.

3.2. Mixture effects of PFAS and heavy metal on cell viability across human cell lines

To evaluate the mixture effects of PFAS and heavy metals on cell viability, HepG2, A498, A549, A431, and PC3 cell lines were treated with varying dose and ratio combinations. In HepG2 and A549 cells, synergistic effects were observed across all ratio combinations, as indicated by significant deviations from the predicted additive response (Fig. 1**a & c)**. In A498 cells, all combinations showed synergistic effects except for the PFOA+As mixtures at 1:3 and 3:1 ratios, which exhibited antagonistic effects (Fig. 1b). Antagonism was also observed in A431 cells for PFOA+Pb at 1:3 and 3:1 ratios, and GenX+Pb at the 1:3 ratio. Similarly, in PC3 cells, the PFOA+As 1:3 combination showed antagonistic interaction. All other mixture combinations demonstrated synergistic effects in both A431 and PC3 cells (Fig. 1d-e**)**. These results highlight the complex and cell type–specific nature of PFAS and heavy metal interactions, with a predominance of synergistic effects on cell viability.

3.3. Different effects of PFAS and heavy metal co-exposure on cell proliferation across human cell lines

The mixture effects of exposure to human-relevant PFAS and heavy metals on cell proliferation were next evaluated using the clonogenic assay, which measured the capacity of individual cells to undergo division and form colonies (N.A. Franken et al., 2006). Unlike the MTT assay, which measures short-term metabolic activity as an indicator of acute cytotoxicity (typically after 24–48 h) (Grela et al., 2018; Kumar et al., 2018; Supino, 1995; Van Meerloo et al., 2011), the colony formation assay captures long-term outcomes, including the capacity of cells to survive treatment and retain proliferative potential (Nicolaas AP Franken et al., 2006; Munshi et al., 2005; Plumb, 2004; Rajendran and Jain, 2017). As such, it provides insight into persistent cytostatic effects that may not be detected in viability assays. Two low-dose concentrations were selected—1 μM PFAS and 0.01 μM heavy metals—to reflect human-relevant exposure levels. The 1 μM PFAS dose (414 μg/L) was chosen to represent the long-term bioaccumulation seen in exposed communities and occupational settings. Human serum levels of PFOA have been reported to range from 1.5 μg/L to 899 μg/L for these two scenarios, corresponding to approximately 3.6 nM to 2.2 μM (ATSDR, 2024). For lead, the WHO's permissible blood concentration is 0.01 mg/L (0.048 μM) (Ali et al., 2024), while occupational exposures have been reported to reach a mean of 0.11 mg/L (0.53 μM) (Mohammadyan et al., 2019). For arsenic, the EPA's limit in drinking water is 0.010 mg/L (0.13 μM) (EPA, 2025). 0.01 μM were selected to approximate human exposure level for heavy metals. The response patterns varied significantly across cell lines. In HepG2 cells, most treatments resulted in a marked reduction in colony formation, except for 1 μM GenX, which showed no significant difference from untreated controls. Notably, lead exposure alone or in combination with 1 μM PFOA led to an over 80 % reduction in colony numbers. While 1 μM GenX alone had no significant effect, its combination with 0.01 μM arsenic resulted in a drastic reduction in colony numbers exceeding 80 %, compared to only a 14 % reduction with arsenic alone (Fig. 2a). This observation indicates non-additive interaction effects. In A549 cells, significant decreases in colony formation were observed with 1 μM PFOA, 0.01 μM arsenic, and 1 μM PFOA combined with 0.01 μM lead (Fig. 2b).

Conversely, in A498, PC3, and A431 cells, certain treatments induced significant increases in colony numbers. In A498 cells, single chemical exposures consistently increased colony formation, and only an increase was observed with the combination of 1 μM PFOA and 0.01 μM lead (Fig. 2c). In PC3 cells, no changes were observed with single exposures, but significant increases occurred with combinations of 1 μM PFOA and 0.01 μM lead, 1 μM PFOA and 0.01 μM arsenic, and 1 μM GenX and 0.01 μM arsenic (Fig. 2d). In A431 cells, significant increases in colony formation were limited to single PFAS exposures, with no mixture effects detected (Fig. 2e). These results showed the cell line-specific nature of the interactions between PFAS and heavy metals, which can either suppress or enhance proliferation depending on the chemical mixtures and the cell type.

3.4. Mixture effects of PFAS and heavy metals on cell migration assessed by scratch assay

To evaluate the impact of PFAS and heavy metal mixture exposures on cell migration and cell-cell interactions, we performed an in vitro scratch assay. In HepG2 cells, significant reductions in wound healing rates were observed following treatment with 1 μM PFOA, 0.01 μM lead, 1 μM PFOA mixed with 0.01 μM arsenic, and 1 μM GenX mixed with 0.01 μM arsenic, compared to untreated controls **(Fig. 3a; Fig. S3)**. A549 cells treated with 0.01 μM lead also exhibited significantly decreased wound healing rates (Fig. 3c). In A431 cells, notable reductions in wound healing were observed across all treatments, including single chemicals and mixtures, except for 1 μM PFOA mixed with 0.01 μM arsenic **(Fig. 3d; Fig. S4)**. No significant changes in wound healing ability were detected in other cell lines under the tested conditions (Fig. 3b & e).

3.5. DNA damage induced by co-exposure to PFAS and heavy metal at human-relevant doses

To evaluate the extent of DNA damage caused by single chemicals and chemical mixtures, the comet assay was conducted on five human cell lines (Fig. 4). All positive controls showed a significant increase in DNA damage, confirming the validity of the assay **(Fig. S5)**. HepG2,

Table 1
Effect concentrations resulting in 10 % reduction in cell viability.

Cell line	Compound	Slope	EC_{10} (μM)	SE(Slope)	SE(EC_{10})
HepG2	**PFOA**	0.12	89.76	0.02	0.17
	GenX	0.20	50.26	0.01	0.03
	Pb	0.20	51.28	0.05	0.12
	As	2.32	4.31	0.27	0.00
A498	**PFOA**	0.32	31.34	0.04	0.04
	GenX	1.16	8.61	0.18	0.01
	Pb	1.69	5.90	0.27	0.01
	As	2.89	3.46	0.58	0.01
A549	**PFOA**	0.06	157.83	0.01	0.19
	GenX	0.09	109.89	0.01	0.14
	Pb	5.82	1.72	0.90	0.00
	As	0.30	33.38	0.07	0.08
A431	**PFOA**	0.19	52.66	0.02	0.05
	GenX	0.25	40.16	0.03	0.04
	Pb	4.67	2.14	0.96	0.00
	As	1.13	8.88	0.30	0.02
PC3	**PFOA**	0.11	89.76	0.01	0.08
	GenX	0.10	103.17	0.01	0.16
	Pb	-0.08	120.45	0.35	5.05
	As	-0.48	20.85	0.09	0.04

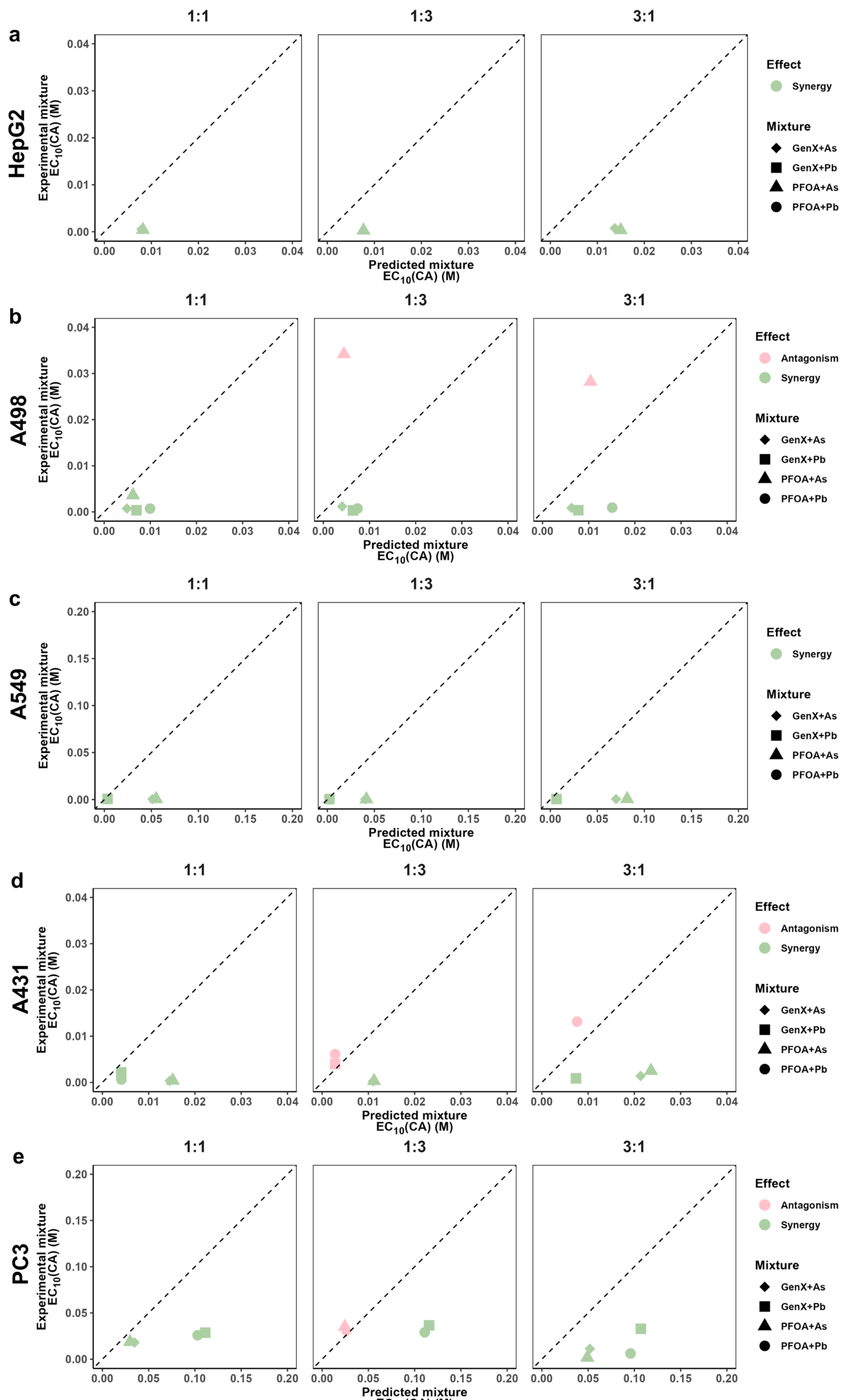

Fig. 1. Comparison between the experimental mixture EC_{10}(CA) with the predicted mixture EC_{10} (CA). (a) HepG2 (b) A549 (c) A498 (d) A431 (e) PC3 cells treated with chemical mixtures at different ratios (1:1, 1:3, and 3:1). Legends indicate the specific mixture treatments: "GenX+Pb" = GenX and Pb mixed at ratios of 1:1, 1:3, and 3:1; "GenX+As" = GenX and As mixed at ratios of 1:1, 1:3, and 3:1; "PFOA+Pb" =PFOA and Pb mixed at ratios of 1:1, 1:3, and 3:1; "PFOA+As" = PFOA and As mixed at ratios of 1:1, 1:3, and 3:1. EC_{10}(CA) indicates EC_{10} of a concentration-additive mixture.

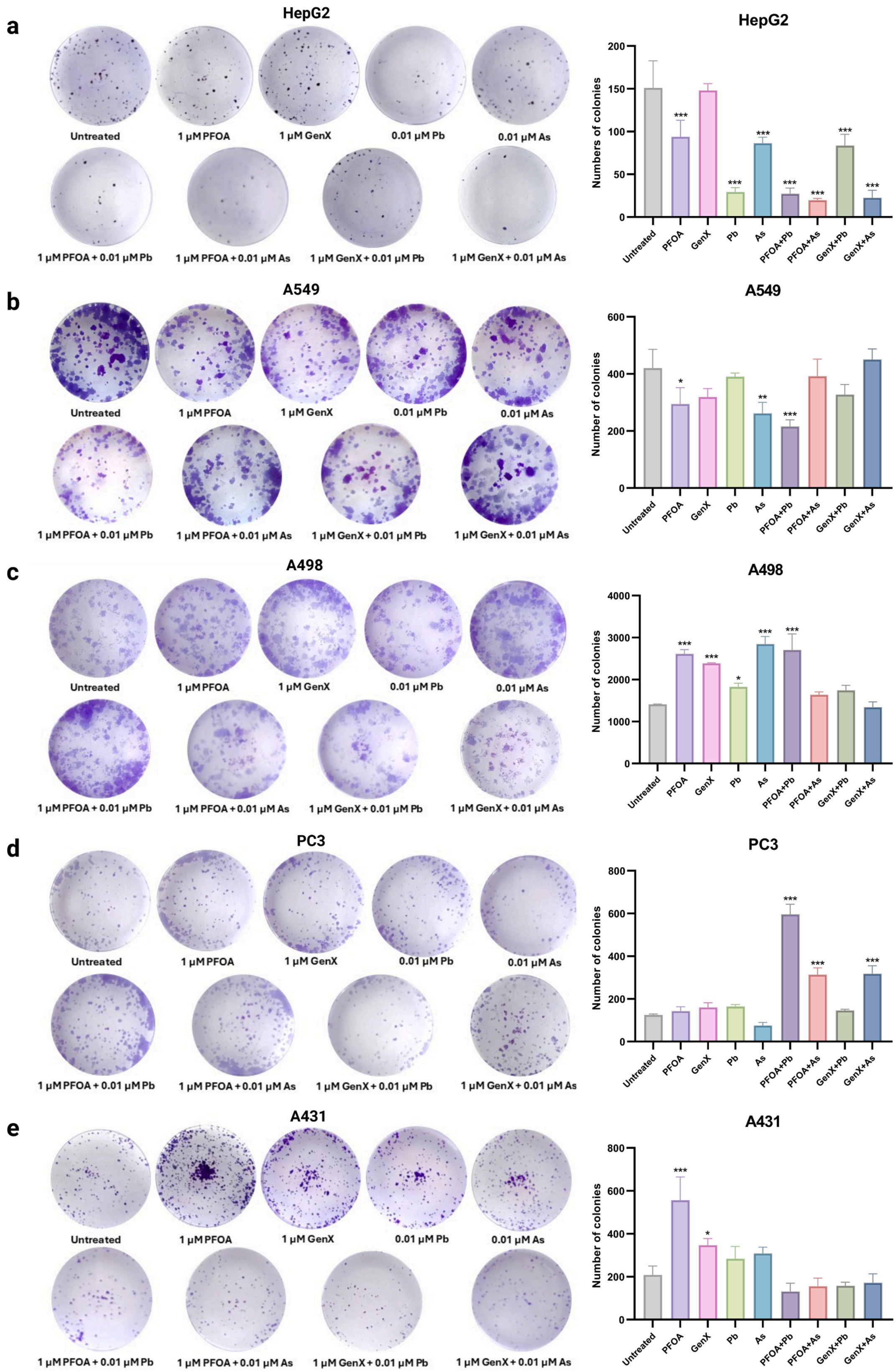

Fig. 2. Cell proliferation of human liver, lung, kidney, prostate, and epidermis cell lines following either single chemical or chemical mixture exposures. Clonogenic assay results are shown for: (a) HepG2, (b) A549, (c) A498, (d) PC3, and (e) A431 cells. Treatments include single exposures to 1 μM PFOA or GenX, 0.01 μM Pb or As, and co-exposures such as 1 μM PFOA + 0.01 μM Pb. Bars in the quantification plots represent the number of colonies formed. Y-axis indicates colony count. X-axis labels correspond to the following concentrations: "PFOA" = 1 μM PFOA, "GenX" = 1 μM GenX, "Pb" = 0.01 μM Pb, "As" = 0.01 μM As, and the same for the chemical mixtures such as PFOA + Pb is corresponding to 1 μM PFOA +0.01 μM Pb. All data are presented as mean ± SD of three independent replicates. Significant codes: $P < 0.05$ (*), $P < 0.01$ (**), $P < 0.001$ (***).

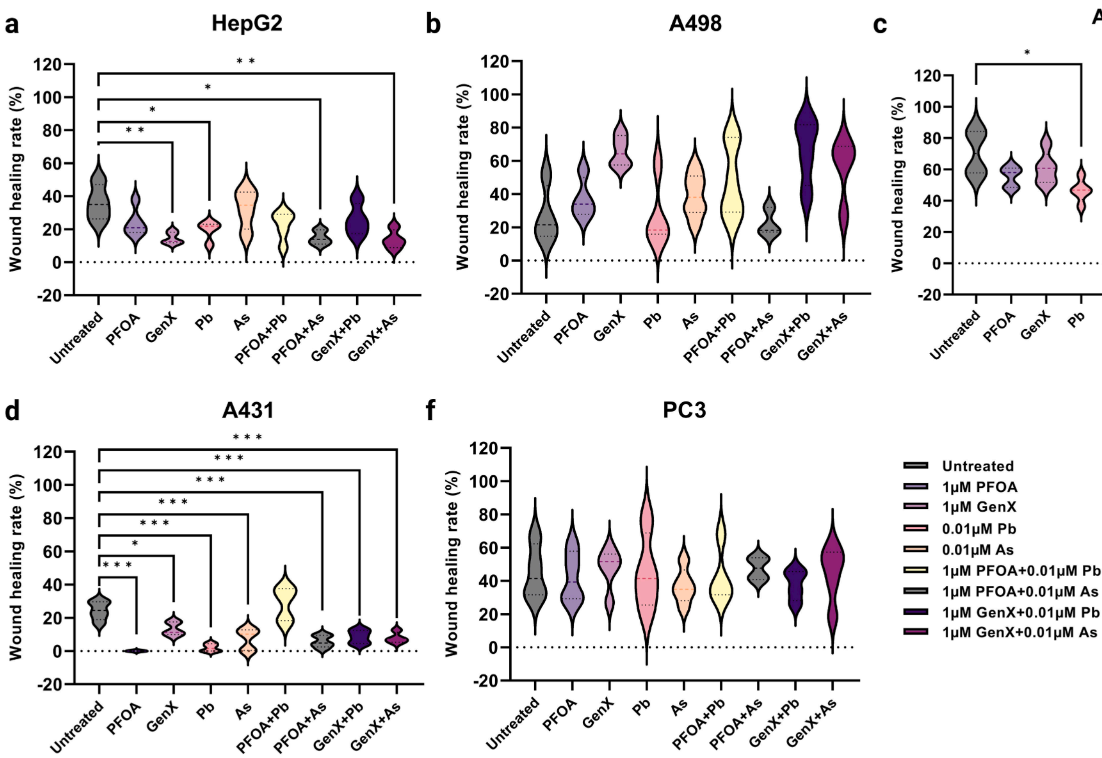

Fig. 3. Cell migration of human liver, kidney, lung, epidermis, and prostate cell lines following either single chemical or chemical mixture exposures. Wound healing ability of (a) HepG2 (b) A498 (c) A498 (d) A549 (e) A431 (f) PC3 cells treated with either single chemical or chemical mixture. All data are presented as mean ± SD of more than four independent replicates. Significant codes: P < 0.05 (*), P < 0.01 (**), P < 0.001 (***).

A549, and A431 cells, exposure to single PFAS, heavy metals, or their combinations resulted in a significant increase in DNA damage compared to the untreated control group. Notably, these three cell lines exhibited a greater degree of DNA damage when exposed to PFAS and heavy metal mixtures compared to individual treatments (Fig. 4a, c, d).

In contrast, A498 cells exhibited significant increase in DNA damage when exposed to 1 μM PFOA, 0.01 μM arsenic, or their combination, compared to untreated controls (Fig. 4b). Similarly, treatment with 0.01 μM lead alone induced marked DNA damage in A431 cells, which was further exacerbated when mixed with single PFAS exposures. Notably, a combination of 1 μM GenX and 0.01 μM arsenic in A431 cells significantly increased DNA damage, even though neither compound alone caused substantial effects. These results indicate both cell type-specific responses and the synergistic potential of PFAS and heavy metal mixtures inducing DNA damage (Fig. 4e).

4. Discussion

Humans are exposed to complex mixtures of environmental contaminants, yet our understanding of the risks posed by persistent chemicals at low exposure levels, especially in combination with other contaminants is limited (Ojo et al., 2021; Rosato et al., 2022; Sinclair et al., 2020). The mechanisms driving the adverse health effects of these mixtures are not well characterized. Epidemiological evidence has identified a significant correlation between PFAS exposure and the presence of certain heavy metals in the U.S. population (Chen et al., 2019; Jain, 2019). However, the combined health impacts of PFAS-heavy metal mixtures and their underlying pathways remain largely unexplored. This knowledge gap is critical, as prior exposure to persistent chemicals may lower human tolerance to additional environmental threats. To address this, our study evaluated the effects of long-chain PFOA, its short-chain alternative GenX, lead, and arsenic, both individually and as mixtures on six human cell lines derived from organs commonly implicated as PFAS targets, including the liver, kidney, lung, skin, and prostate (Lucas et al., 2023; Rafiee et al., 2024; Seyyedsalehi and Boffetta, 2023; Wen et al., 2020; Zhang et al., 2022). By examining these co-exposures, important insights into the risks associated with PFAS and heavy metal mixtures could be garnered, contributing to a more comprehensive understanding of their impact on human health.

PFAS, recognized as endocrine disruptors, impact human organs at varying levels of severity. Human exposure to PFAS occurs through multiple sources and routes. PFAS can leach from consumer products such as paper packaging and nonstick cookware and agricultural produce. Additionally, contamination of soil, groundwater, and air by nearby industries serves as a significant source of environmental exposure (Glüge et al., 2020; Trudel et al., 2008). Occupational exposure further compounds these risks; for example, firefighters experience direct skin exposure from contact with firefighting foam and protective clothing (Lucas et al., 2023). Dermal absorption of PFAS has been associated with immunotoxicity and may lead to allergic reactions in humans (Franko et al., 2012; Han et al., 2020; Shane et al., 2020). Similarly, inhalation exposure can result in inflammation, impaired lung function, and an elevated risk of lung cancer (Barry et al., 2013; Dragon et al., 2023; Qin et al., 2017). Once PFAS enters the body, it tends to accumulate in the liver and kidneys, where it can induce hepatotoxicity and nephrotoxicity (Bartell and Vieira, 2021; Li et al., 2017; Wen et al., 2020; Zhang et al., 2022). Moreover, PFAS exposure has been linked to reproductive toxicity, including the promotion of prostate cancer (Tarapore and Ouyang, 2021; Wei et al., 2023). To model tissue-specific responses, we selected human cell lines representative of organs commonly affected by PFAS and heavy metal exposure. HepG2 (liver) and A498 (kidney) were chosen due to the key roles of these organs in xenobiotic metabolism and excretion. A549 (lung) and A431 (skin) represent inhalation and dermal exposure pathways. PC3 (prostate) was selected due to increasing evidence of PFAS-related reproductive toxicity. Together, these cell lines provide a robust platform for evaluating tissue-specific toxicity.

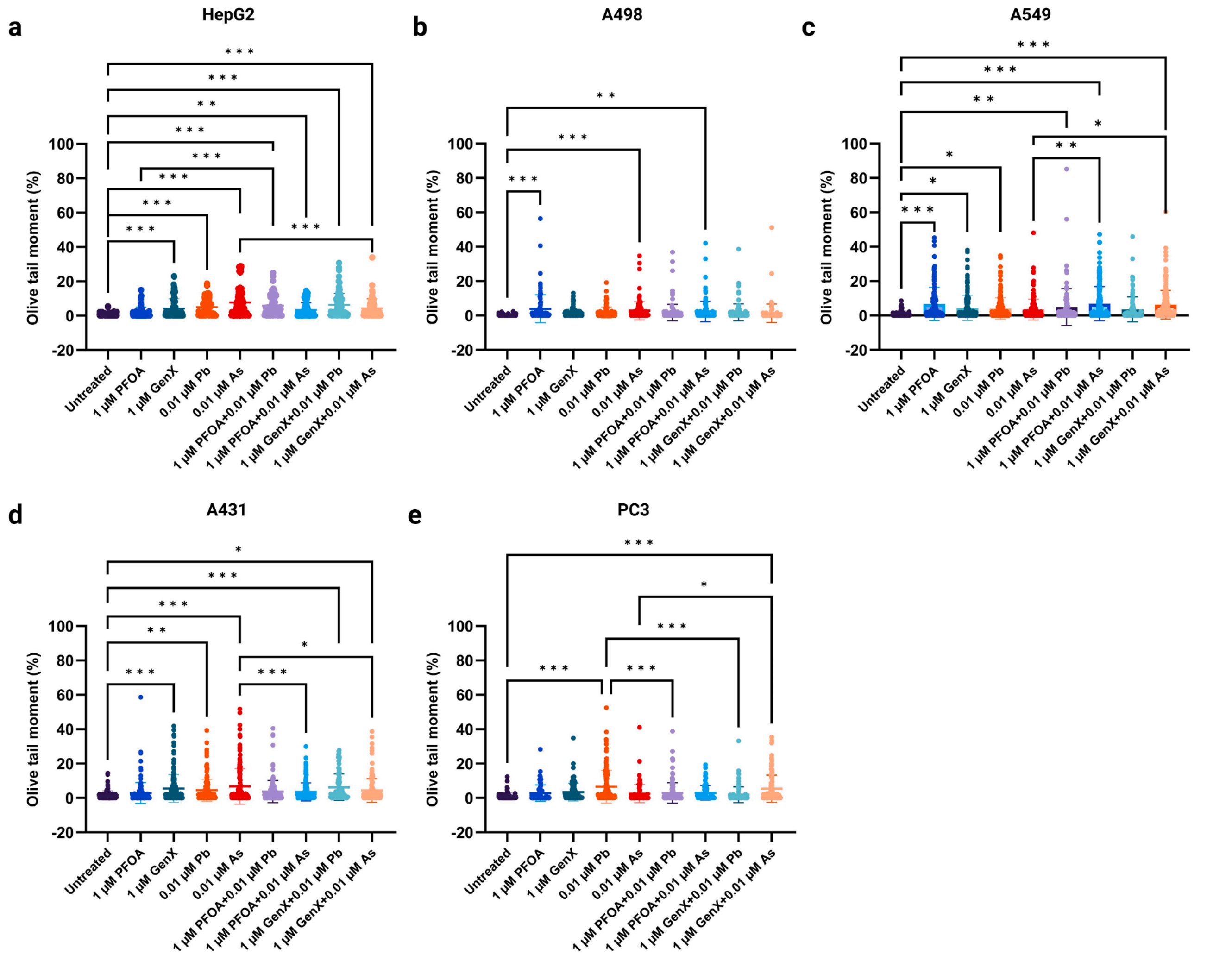

Fig. 4. Effects of single and PFAS and heavy metal mixture exposures on DNA damage in human cell lines assessed by the comet assay. For each treatment, 100–200 comets were scored across at least two biological replicates. Experiments were repeated twice, yielding consistent results. Statistical significance indicated as follows: ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$.

In our study, the effects of PFOA and its short-chain alternative GenX on cell viability were evaluated using six human cell lines derived from the liver, kidney, lung, skin, and prostate tissues to represent a comprehensive investigation of PFAS and heavy metal mixture effects in vitro. Traditionally, short-chain PFAS such as GenX are expected to be less toxic than their long-chain counterparts due to their lower bioaccumulation potential (Li et al., 2020a). However, our findings indicate that GenX can exert cytotoxic effects compared to PFOA in certain cell lines and at specific exposure concentrations. This may be explained by the fact that, as a short-chain alternative to PFOA, GenX shares similar physicochemical properties, including the presence of strongly polar carboxylate groups, which can drive comparable interactions with cellular components (Shi et al., 2024; Wen et al., 2020). These results challenge the assumption that short-chain PFAS like GenX are safer alternatives. Previous studies have similarly raised concerns regarding the toxicity and environmental persistence of short-chain PFAS (Ateia et al., 2019; Brendel et al., 2018; Solan et al., 2023) Given their potential to induce biological effects comparable to long-chain PFAS, our results highlight the need for more stringent regulatory evaluation of PFAS substitutes. Replacement compounds should be not only less persistent in the environment but also demonstrably less toxic to human health. Our observation of dose-dependent cytotoxicity in HepG2 cells aligns with previous reports were consistent with prior studies (Wen et al., 2020). However, in A549 lung cells, a decrease in cell viability to approximately 80 % was noted at 240 μM GenX exposure, whereas earlier research reported increased cell viability at 200 μM GenX (Jabeen et al., 2020). This discrepancy may stem from differences in the culture media used; our study cultured the cells with DMEM, while the previous study utilized F-12 K medium. Both media are suitable for A549 cell culture but may yield differing long-term effects (Cooper et al., 2016). Single heavy metal exposures demonstrated dose-dependent decreases in cell viability across liver, kidney, lung, and skin cell lines, while the prostate cell lines showed an increase in viability. These observations align with prior studies on how lead and arsenic influenced cell viability (Choi et al., 2018; Gutowska et al., 2016; Naranmandura et al., 2007; Wei et al., 2023; Yedjou and Tchounwou, 2007).

To date, the synergistic impact of PFAS and other environmental toxins has become a significant area of concern due to the simultaneous exposure of individuals to multiple environmental contaminants. While the acute toxicity of PFAS and heavy metals as individual compounds has been extensively studied in human cell lines and animal models, their combined impact remains largely unexplored, despite their frequent co-occurrence in the U.S. population (Chen et al., 2019; Jain, 2019). Moreover, the underlying mechanisms of these interactions are not well understood. In this study, we focused on low-effect levels (<30 %) and used a linear model for EC10 calculation, excluding any effects above 30 %, as this approach is typically suitable for dose-response relationships at these levels (Escher et al., 2020, 2018; Ríos-Bonilla et al., 2024). Our primary aim was to capture environmentally relevant exposure levels. However, for effects greater than 30 %, a 3- or 4-parameter logistic model would be more appropriate for establishing EC values such as EC50.

Our evaluation of PFAS-heavy metal mixture effects showed that the expected mixture interaction depends on both chemical combination and dose-ratio. Synergistic effects were observed across all six cell lines at different dose ratios. Specifically, for HepG2 and A549 cell lines, the observation of synergistic effects under all three dose ratios suggests that the combined presence of PFAS and heavy metals enhances cellular toxicity beyond what would be expected from their individual effects, regardless of the proportion in the mixture. This indicates that their interactions are not simply additive but may involve shared or converging toxicological pathways. The antagonistic effects observed in A498 kidney cells when treated with PFOA + arsenic at both 1:3 and 3:1 dose ratio may suggest that the combined exposure reduces the overall toxicity compared to what would be expected from each chemical alone. One possible explanation could be PFOA and arsenic may interfere with each other's uptake, distribution, or intracellular binding sites (Cedergreen, 2014; Hernández et al., 2017). Similar antagonistic interactions were observed in PC3 cells with PFOA/GenX + arsenic at a 1:3 ratio, and in A431 skin cells with PFOA/GenX + lead at a 1:3 ratio as well as with PFOA + lead at a 3:1 ratio. These findings highlight the complexity of mixture toxicology, where both the specific chemical pairings and their relative proportions can significantly alter cellular responses.

In this study, 1 μM PFOA was selected as the human-relevant dose, which corresponds to 414.07 ng/mL of PFOA, which is within the exposure range of affected communities, particularly those with occupational exposure (ATSDR, 2024). the same concentration was used for GenX. For heavy metals, 0.01 μM was chosen as the human-relevant dose, equivalent to 0.0033 mg/L for lead and 0.0013 ng/mL for arsenic, levels significantly lower than the WHO's permissible blood concentrations (Ali et al., 2024; Gorchev and Ozolins, 1984). Cell proliferation and DNA damage data revealed cell line-specific interactions between PFAS and heavy metals, with varying effects based on the cell type and chemical combination. Interestingly, a correlation between cell proliferation, migration, and DNA damage was observed in several cell lines. In HepG2 liver cells, an overall impaired proliferation was found under both single chemical and PFAS and heavy metal mixture exposure. Correspondingly, DNA damage was significantly elevated, and wound healing ability was impaired, suggesting that the mixture effects of PFAS and heavy metals may primarily impair cell function via DNA damage pathways compared to untreated controls. A similar trend was observed in A549 lung cells, where proliferation decreased, DNA damage increased, and wound healing was modestly affected compared to untreated controls. We hypothesize that the observed impairment in cell proliferation, migration, and DNA integrity could be driven by mechanisms such as oxidative stress, DNA damage response pathways, and inflammation. PFAS and heavy metals are known to generate reactive oxygen species (ROS), which can induce cellular oxidative stress, leading to DNA damage and disruption of vital cellular processes (Coperchini et al., 2020; Liu et al., 2007; Moura et al., 2011; Wielsøe et al., 2015; Zhang et al., 2024). Additionally, both PFAS and heavy metals can disrupt various signaling pathways involved in cell cycle regulation, apoptosis, and repair mechanisms (Pulido and Parrish, 2003; Rana, 2008; Robarts et al., 2022; Wen et al., 2020), potentially contributing to impaired cell proliferation and migration. Future studies investigating the specific signaling pathways and oxidative stress markers in response to PFAS and heavy metal co-exposure would be crucial for better understanding the molecular mechanisms at play.

In contrast, A498 kidney and PC3 prostate cell lines exhibited an increasing trend in proliferation under PFAS and heavy metal co-exposure, with no significant changes in wound healing rates and less pronounced DNA damage compared to other cell lines. These findings suggest that, in kidney and prostate cells, PFAS and heavy metals may affect cellular viability through different toxicological mechanisms other than DNA damage pathways. For A431 skin cells, significant DNA damage and a pronounced reduction in wound healing ability were observed across most single and mixture exposures. However, no significant changes in proliferation rates were detected under mixture treatments, indicating that PFAS and heavy metals may exert distinct toxicological effects on skin cells. These findings highlight the importance of considering co-exposure scenarios in risk assessment, since even low levels of contaminants may produce disproportionately large biological effects when combined.

While our study offers important insights into the synergistic toxic effects of PFAS and heavy metal mixtures across multiple human cell lines, it is important to acknowledge the limitations of in vitro models. These systems do not fully replicate the complexity of in vivo environments, such as the influence of metabolism, tissue-tissue interactions, immune responses, and systemic clearance mechanisms (Devlin et al., 2005; Ghallab and Bolt, 2014; Greim et al., 1986). For example, compounds like PFAS undergo hepatic biotransformation in vivo (Kolanczyk et al., 2023), which may alter their toxicity profiles. Additionally, the cellular microenvironment in tissues may influence chemical uptake and response, which cannot be entirely captured in monoculture systems. Therefore, while our findings provide a useful foundation, further studies using animal models and human-relevant organotypic systems will be essential to validate the observed effects and better understand their implications for human health.

5. Conclusions

This study highlights the effects of PFAS and heavy metals, individually and in combination, on human cell viability, proliferation, migration, and DNA damage, underscoring the need to understand their synergistic and antagonistic interactions given widespread environmental and occupational exposures. Synergistic and antagonistic effects observed varied depending on cell type, chemical combination, and dose ratio, with clear cell line-specific responses: such as HepG2 and A549, demonstrating impaired proliferation and viability due to DNA damage under mixture exposure, while others, such as A498 and PC3, exhibited increased proliferation and minimal DNA damage. These divergent outcomes suggest that the toxicological impact of PFAS and heavy metals is tissue-specific and mediated by distinct mechanisms depending on the chemical combinations. Importantly, our findings emphasize that the effects of chemical co-exposures cannot be reliably predicted from single-compound toxicity data, warranting the need for more nuanced risk assessments.

We acknowledge several limitations in this study. First, while in vitro models provide valuable mechanistic insights, they do not fully recapitulate the complexity of human physiology, including metabolism, immune responses, and tissue-level interactions. Second, our focus on acute responses does not capture the effects of chronic, low-dose exposures that are more representative of environmental scenarios. Third, the study was limited to a small subset of PFAS and heavy metals, and expanding the chemical scope would be essential for broader relevance. To address these gaps, future research should incorporate in vivo studies to better reflect systemic biological processes and long-term health outcomes. Additionally, omics-based approaches, such as transcriptomics and proteomics, could help identify disrupted molecular pathways, including those related to oxidative stress, DNA repair, and cell cycle control. Such mechanistic insights would aid in identifying biomarkers of exposure and toxicity. Ultimately, integrating data from in vitro, in vivo, and omics studies within a systems toxicology framework will be critical for advancing chemical safety evaluations and guiding the development of safer alternatives.

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CRediT authorship contribution statement

Mia Sands: Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Arshveer Sachdeva:** Software, Data curation. **Laura Bukavina:** Writing – review & editing, Methodology. **Joseph Irudayaraj:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare no competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.hazadv.2025.100913.

Data availability

Data will be made available on request.

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